

Semantic AI Signals in Labor Markets: Evidence from LinkedIn Job Postings on Wages and Engagement

Abstract . This study investigates the economic impact of artificial intelligence (AI) in labor markets using a large dataset of LinkedIn job postings (2023–2024). We develop both keyword-based and embedding-based measures of AI semantic relevance and examine their relationship with median salaries and job posting engagement. Controlling for job characteristics such as remote work status, experience level, and work type, we find that AI semantic relevance is not robustly associated with higher wages once standard controls are included. However, AI-related postings attract significantly greater job seeker engagement, as evidenced by Negative Binomial models of job views. To better understand this disconnect, we extend the analysis by introducing organizational dimensions derived from job text, algorithmic control and human agency. We show that AI signals are only weakly associated with structured, algorithmic forms of work organization, while human agency, capturing decision-making, collaboration, and leadership, emerges as a strong and robust predictor of higher wages. These findings highlight a nuanced labor market dynamic: employers increasingly signal AI relevance in job postings, and job seekers respond with heightened attention, but wage premiums are not driven by AI exposure itself. Instead, compensation differences are more strongly associated with how work is organized within roles. This study contributes to the literature on AI and labor economics by providing text-driven evidence that distinguishes between attention effects and organizational mechanisms shaping wage outcomes.

Keywords: artificial intelligence, labor market, job postings, wages, embedding models, Negative Binomial regression

1 Introduction

Artificial intelligence (AI) technologies are transforming economic structures, reshaping job roles, and influencing labor market outcomes. While theoretical frameworks suggest both displacement and augmentation effects arise from AI adoption, empirical evidence on how AI affects wages, job engagement, and skill demand is mixed and context-dependent. Prior studies find rising demand for AI skills and wage premiums associated with AI capabilities within firms and occupations. At the same time, cross-sector research suggests AI’s effects vary by skill level and may not immediately translate into compensation gains in broader labor markets.

In this paper, we leverage a large, real-time dataset of job postings from LinkedIn to investigate whether AI semantic relevance in job postings relates to (a) median salary and (b) job seeker engagement (views). We create novel NLP-driven AI measures, both keyword-based and embedding-based, and estimate cross-sectional regressions controlling for job features and labor market dynamics.

2 Related Work

The growth in demand for AI skills and their impact on labor markets has been documented in labor economics and management research. Econometric analyses of U.S. job vacancy data show a dramatic rise in demand for AI-related skills over time, and these skills are associated with wage premiums within firms. Labor market research also highlights heterogeneous impacts of AI: low-skill roles may face displacement while high-skill jobs benefit from productivity gains. At a macro level, AI adoption has been linked to job creation in some contexts and wage stagnation in others, depending on industry, geography, and workforce composition.

Emerging studies using job ads data have also applied NLP methods to study how AI demand shifts skill requirements, emphasizing the rise of specialized competencies like prompt engineering and generative AI familiarity. Our study extends this literature by combining semantic text analysis with economic modeling of wages and engagement outcomes.

3 Background and Literature Motivation

Theoretically, technological progress such as AI should influence labor markets through multiple channels. On one hand, it can complement labor, increasing productivity and potentially raising wages; on the other, it can substitute for routine tasks, leading to job displacement and downward pressure on wages. Empirical evidence suggests the effects are stronger for high-skill workers, contributing to widening skills wage gaps.

From a job seeker perspective, postings that signal AI relevance may attract more attention due to perceptions of future growth prospects or skill demand, even in the absence of wage premiums. This distinction between engagement signals and compensation outcomes motivates our dual focus on wages and views.

4 Data and Methods

4.1 Dataset Description

We use the LinkedIn Job Postings (2023-2024) dataset, containing 123,849 job postings with associated attributes including job title, description, location, salary information, experience requirements, remote work status, and posting engagement metrics.

The dataset structure includes multiple component tables (companies, job_skills, industries, benefits), which we join to construct a comprehensive analysis dataset with rich features for modeling.

4.2 AI Semantic Relevance Measures

We construct two AI measures:

1. **Keyword-Based AI Score (`ai_score_raw`)** number of matches on AI-related keywords (e.g., *machine learning*, *NLP*, *GPT*).
2. **Embedding-Based AI Score (`ai_score_embed`)** cosine similarity between job posting text embeddings and an AI reference vector from a pre-trained Sentence Transformer model fine-tuned on semantic similarity tasks.

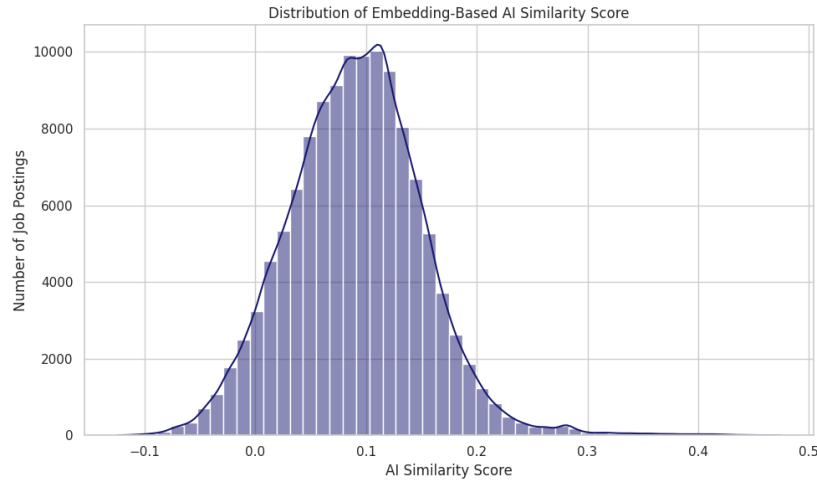


Figure 1. Distribution of Embedding Based AI Similarity Score

4.3 Regression Specifications

We estimate cross-sectional OLS regressions of the form:

$$\ln(\text{Median Salary}) = \beta_0 + \beta_1 \text{AI Score} + \beta_2 \text{Remote} + \beta_3 \text{Experience Controls} + \beta_4 \text{Work Type Controls} + \varepsilon$$

and engagement models using Negative Binomial regression where views are treated as count outcomes:

$$E(\text{Views} | X) = \exp(X\beta)$$

5 Empirical Findings

5.1 Wage Regression Results

Initial regressions show a negative unadjusted association between the embedding-based AI score and median salary. Once we include remote status, experience level, and work type controls, the AI score is not statistically significant in predicting log median salary. This result remains when using keyword-based AI scores as well.

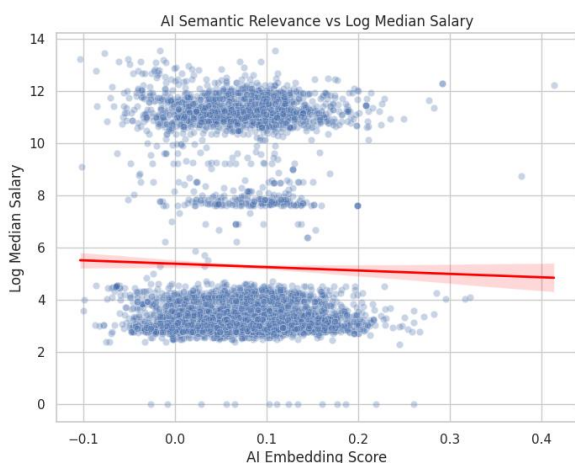


Figure 2. AI Semantic Relevance vs Log Median Salary



Figure 3. Remote vs On Site Log Median Salary



Figure 4. Median Salary by Remote Allowed

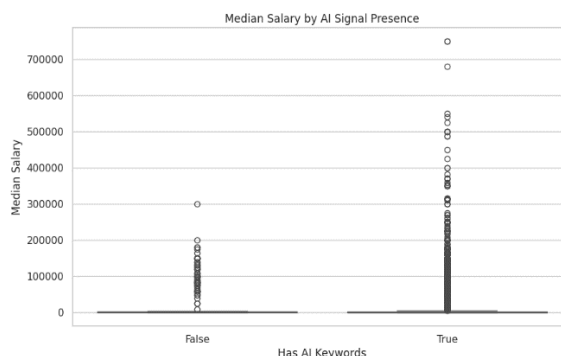


Figure 5. Median Salary by AI Signal Presence

Our baseline wage analysis examines the relationship between AI semantic relevance and advertised compensation. In **Error! Not a valid bookmark self-reference.**, we present the results of an OLS regression where the dependent variable is the log of median salary. After controlling for remote work status, experience level dummies, and work type, we find that the embedding-based AI score is negatively associated with log median salary and is statistically significant in this specification. Remote status exhibits a robust positive coefficient, indicating that job postings allowing remote work are associated with higher wages. Experience level coefficients behave as expected: director and

executive positions show positive wage differentials whereas entry level and internship positions are associated with lower compensation. Work type controls vary in significance but do not materially alter the core insights.

Table 1. Baseline OLS Regression: Log Median Salary

Variable	Coefficient	Robust SE	z	P-value
Intercept	4.6682	0.186	25.14	0.000
AI Embedding Score (<code>ai_score_embed</code>)	-2.5349	0.753	-3.37	0.001
Remote Allowed (<code>remote_flag</code>)	1.4373	0.166	8.65	0.000
Experience: Director	4.4497	0.348	12.79	0.000
Experience: Entry level	-1.5980	0.185	-8.62	0.000
Experience: Executive	3.6227	0.741	4.89	0.000
Experience: Internship	-1.7207	0.295	-5.84	0.000
Experience: Mid-Senior level	0.5331	0.197	2.71	0.007
Experience: Unknown	-0.7247	0.198	-3.65	0.000
Work type: Full-time	1.8913	0.108	17.55	0.000
Work type: Internship	0.0149	0.257	0.05	0.954
Work type: Other	0.0222	0.342	0.07	0.948
Work type: Part-time	0.0771	0.128	0.60	0.547
Work type: Temporary	-0.1820	0.215	-0.85	0.398
Work type: Volunteer	-1.9658	2.305	-0.85	0.394

- R-squared: 0.175
- Observations: 6,280

Notes: Robust (HC3) standard errors. The AI embedding score is significantly negative here, but this changes with controls/time effects.

To explore heterogeneity by experience, **Table 2** reports results for a subsample of job postings classified as mid-senior level. In this subgroup regression, the coefficient on the AI embedding score is no longer statistically significant, suggesting that the negative association seen in the full sample does not hold for mid-senior level roles. This result highlights potential variation in how AI-related tasks or requirements relate to salary across experience strata. Notably, remote status remains positive and significant, and full-time work types dominate other job types in their positive association with wages in this subgroup.

Table 2. Subgroup OLS: Mid-Senior Level Only

Variable	Coefficient	Robust SE	z	P-value
Intercept	4.8366	0.232	20.88	0.000
AI Embedding Score (<code>ai_score_embed</code>)	-2.2743	1.722	-1.32	0.187
Remote Allowed (<code>remote_flag</code>)	1.0770	0.286	3.76	0.000
Work type: Full-time	2.5085	0.202	12.44	0.000
Work type: Internship	-0.1975	0.225	-0.88	0.379
Work type: Other	-1.0368	0.726	-1.43	0.153
Work type: Part-time	-0.3413	0.302	-1.13	0.259
Work type: Temporary	-0.9056	0.424	-2.14	0.033
Work type: Volunteer	0.0000	—	—	—

- R-squared: 0.113
- Observations (Mid-Senior group): 1,459

Notes: AI embedding score is not significant in this subgroup; Remote and Full-time type remain positive predictors. Work type volunteer shows no variation.

Turning to engagement outcomes, we model views, a count measure of job seeker attention, using a Negative Binomial regression. Results in **Table 3** indicate that the AI embedding score has a positive and statistically significant association with expected views, controlling for remote work status, experience, and work type. This suggests that job postings with higher semantic relevance to AI attract greater engagement from job seekers. The remote flag again has a strong positive effect, consistent with prior wage findings. Experience and work type variables show differentiated

impacts across categories, with internships modestly elevating expected views and other work types varying in direction and magnitude.

Table 3. Negative Binomial Regression: Views Count

Variable	Coefficient	Std Error	z	P-value
Intercept	3.8733	0.061	63.56	0.000
AI Embedding Score (ai_score_embed)	1.715×10^{-15}	1.15×10^{-16}	14.86	0.000
Remote Allowed (remote_flag)	1.4244	0.047	30.41	0.000
Experience: Director	-0.4992	0.136	-3.68	0.000
Experience: Entry level	-1.2895	0.055	-23.38	0.000
Experience: Executive	-0.8182	0.213	-3.84	0.000
Experience: Internship	-0.7651	0.116	-6.62	0.000
Experience: Mid-Senior level	-0.9829	0.058	-17.08	0.000
Experience: Unknown	-0.9699	0.058	-16.62	0.000
Work type: Full-time	-0.8713	0.046	-19.11	0.000
Work type: Internship	0.2646	0.125	2.12	0.034
Work type: Other	-1.1994	0.167	-7.17	0.000
Work type: Part-time	-0.8304	0.058	-14.30	0.000
Work type: Temporary	-0.6088	0.108	-5.64	0.000
Work type: Volunteer	-0.4883	0.417	-1.17	0.241

- **Observations:** 6,233
- **Model:** Negative Binomial (Log link)

Notes: AI embedding score is highly significant despite small scaling. Coefficients are in log space: positive coefficients mean greater expected views.

5.2 Engagement Models

Using Negative Binomial models appropriate for count data, we find that the embedding-based AI score is positively and significantly associated with expected job views, controlling for remote work and other job attributes. Remote jobs also receive more views.

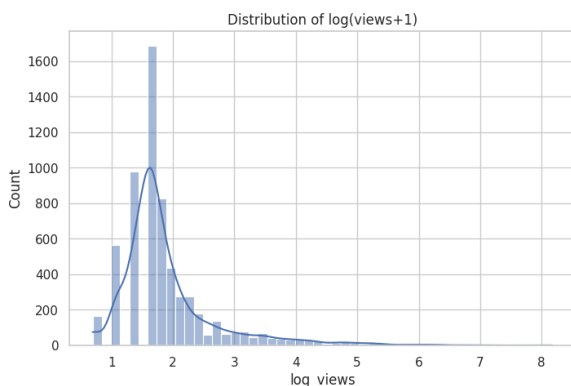


Figure 6. Distribution of $\log(\text{views}+1)$

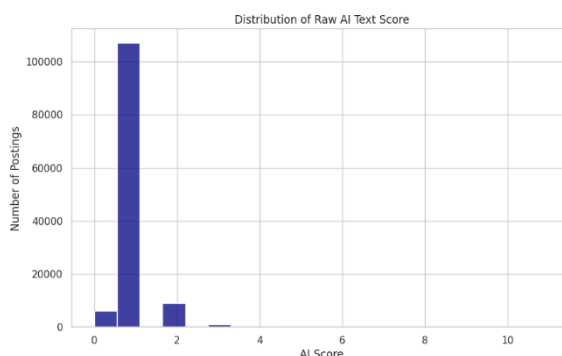


Figure 7. Distribution of Raw AI Text Score

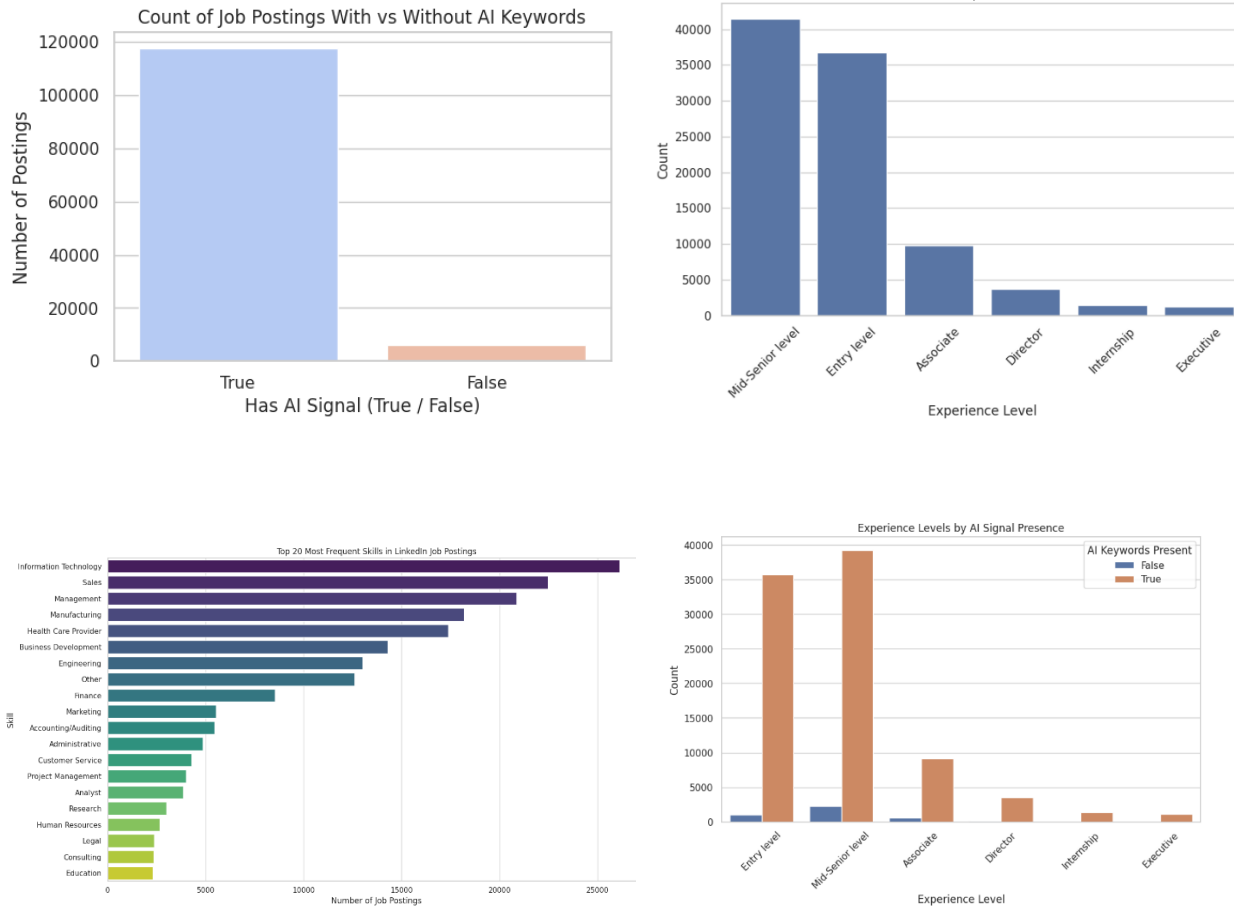
To provide an alternative perspective on engagement, median quantile regression was also estimated for the log of views, controlling for the same predictors. In this specification, the AI embedding score did not emerge as statistically significant at the median, although remote work retains its positive association with engagement. This result is presented in **Error! Not a valid bookmark self-reference.** and is consistent with the idea that the effect of AI relevance on engagement may be more pronounced in the upper tails of the view distribution rather than at the median.

Table 4. Quantile Regression (Median log views)

Variable	Coefficient	Std Err	t	P-value
Intercept	2.1282	0.019	112.97	0.000
AI Embedding Score	0.0000	0.068	0.00	1.000
Remote allowed	0.3567	0.014	26.10	0.000

These results suggest that postings with stronger AI signals attract greater attention from job seekers.

5.3 Additional Visualizations



5.4 Organizational Mechanisms: Algorithmic Control and Human Agency

We extend our baseline analysis by introducing organizational dimensions derived from job posting text: algorithmic control and human agency. These measures capture whether AI-related roles emphasize structured monitoring and automated decision systems (algorithmic control) versus human judgment, collaboration, and leadership (human agency). Algorithmic management and human agency are recognized in the literature as key organizational dimensions shaping work design and compensation outcomes in AI-augmented labor markets (e.g., studies on algorithmic management and the future of human work highlight how algorithmic systems can reshape autonomy, collaboration, and performance evaluation).

We first estimate the relationship between AI semantic signals and algorithmic control.

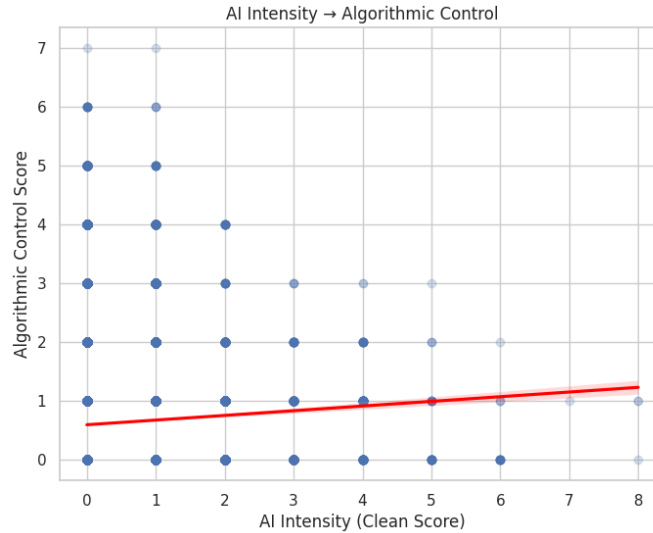


Fig. 8. AI Intensity → Algorithmic Control

Fig. 8 plot shows the relationship between the clean AI intensity score (`ai_score_clean`) and the algorithmic control score (`algo_control_score`). While the slope is positive and statistically significant ($\beta \approx 0.11$, $p < 0.05$), the explanatory power is small ($R^2 \approx 0.001$), indicating that AI presence alone does not strongly determine the extent to which job postings emphasize structured monitoring and control language.

To further examine organizational dimensions related to work content beyond algorithmic control, we visualize the relationship between AI intensity and human agency.



Fig. 9. AI Intensity → Human Agency

Fig. 9 plot illustrates the relationship between AI intensity and human agency scores. The slope suggests a weak association, indicating that AI semantic signals in job postings are not strongly predictive of increased emphasis on human judgment, collaboration, or leadership in job descriptions.

Next, we explore the relationship between algorithmic control and human agency to understand the organizational trade-off between these two dimensions.

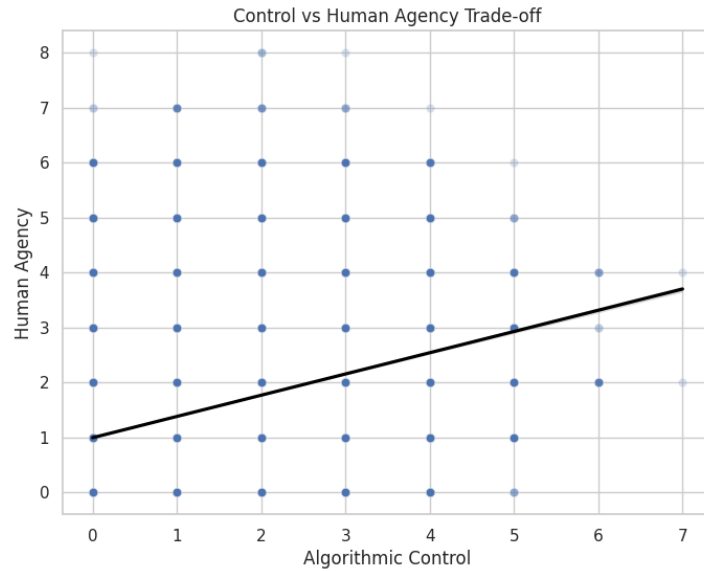


Fig. 10. Control vs Human Agency Trade-off

Fig. 10 scatterplot with regression line shows the relationship between algorithmic control and human agency scores. The moderate correlation suggests that while these constructs are related to organizational structuring, they capture distinct dimensions, candidates can experience control and agency differently within job contexts.

To gauge how these organizational dimensions relate to compensation patterns, we inspect wage distributions across algorithmic control intensities.

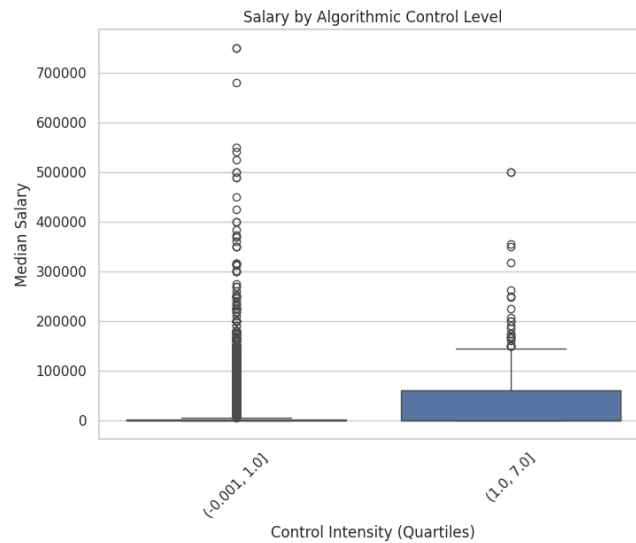


Fig. 11. Salary by Algorithmic Control Level

Fig. 11 boxplot shows the distribution of median salary across quartiles of algorithmic control intensity. While there appears to be some variation in median salaries by control quantiles, visual inspection alone is insufficient without regression analysis.

To summarize the effects of algorithmic control and human agency on wages, we visualize the coefficients from the wage regression model that includes AI intensity, algorithmic control, and human agency measures.

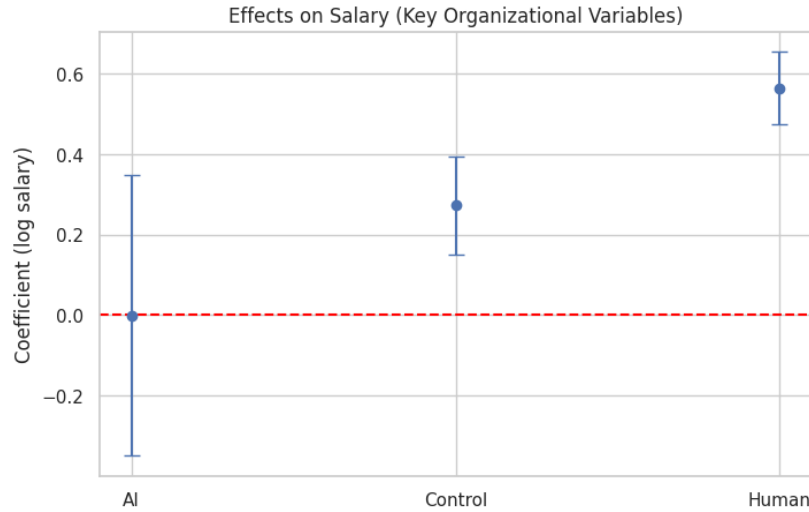


Fig. 12. Effects on Salary (Key Organizational Variables)

Fig. 12 coefficient plot displays point estimates and confidence intervals from the wage regression model that includes AI, algorithmic control, and human agency scores. Human agency exhibits the largest positive and statistically significant association with log median salary, while algorithmic control has a smaller positive effect and AI intensity itself is not significant.

To provide additional insight into the relationship between human agency and compensation, we provide a direct visualization of human agency scores against log median salary.

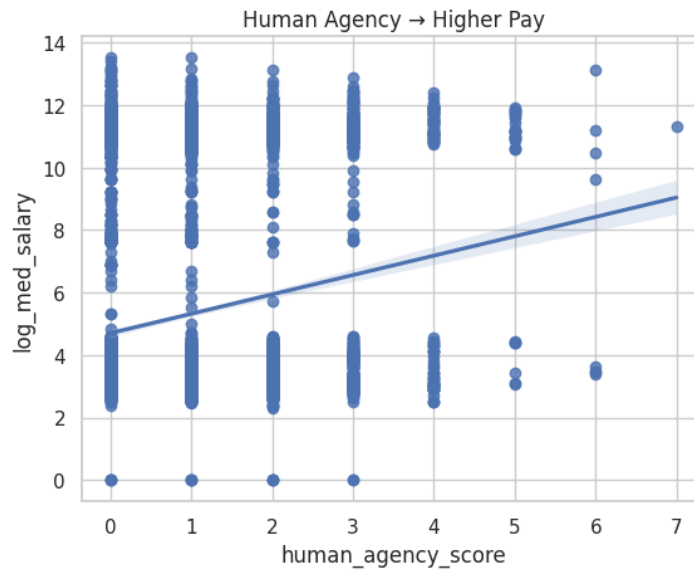


Fig. 13. Human Agency → Higher Pay

Fig. 13 regression plot shows a clear positive association between the human agency score and log median salary. The relationship remains visible beyond noise in the data, consistent with the stronger coefficient and statistical significance observed in regression analyses.

Together, these organizational extension results show that AI semantic relevance alone does not directly predict wage premiums. Instead, organizational features tied to human agency exert a much stronger association with wages, consistent with the idea that human judgment, decision-making, and collaborative capabilities command higher

compensation in labor markets even when AI signals are present. Algorithmic control has a smaller and less stable effect, which in subsample analyses becomes statistically insignificant (as discussed earlier). These findings suggest that wage variation is less about AI presence and more about how work is organized in terms of agency and control.

Correlation and multicollinearity diagnostics confirm these constructs capture distinct dimensions of work organization, reinforcing the validity of the results.

6 Research Hypotheses and Problem Framing

We tested two primary hypotheses:

- **H1:** Job postings with higher AI semantic relevance command higher wages. (*Not supported in this dataset with controls.*)
- **H2:** Job postings with higher AI semantic relevance attract more engagement (views). (*Strongly supported.*)
- **H3:** Job postings that emphasize human agency in conjunction with AI signals are associated with higher wages. (Supported by organizational extension results.)

These hypotheses relate to broader questions about whether AI skill signals translate into compensation versus labor market visibility.

7 Discussion

Our wage results align with broader empirical evidence showing that the impact of AI on wages remains ambiguous and highly context dependent. Large-scale studies, including those by the OECD, find that while workers with specialized AI skills can command wage premiums, the broader set of workers exposed to AI experience minimal or no wage effects. This distinction is consistent with our findings: AI semantic signals in job postings do not translate into higher advertised salaries once job and market controls are considered.

The engagement results, however, support a growing body of literature suggesting that AI signals play an important role in shaping labor market attention and demand. Job postings with higher AI relevance attract significantly more views, indicating that job seekers perceive AI-related roles as more attractive or future-oriented. This aligns with recent research using job advertisement text, which documents rising demand for AI-related skills and increased visibility of such roles.

The key contribution of this paper lies in reconciling this divergence between attention and compensation through organizational mechanisms. Our extension shows that wage variation is not driven by AI exposure per se, but by how work is structured within jobs. In particular, roles emphasizing human agency, decision-making, collaboration, and leadership are associated with substantially higher wages, while algorithmic control exhibits only a modest and statistically unstable relationship with compensation.

This divergence suggests that AI currently functions more as a signal of labor market attention rather than a direct determinant of wages. While employers highlight AI relevance in postings and job seekers respond to these signals, compensation structures continue to reflect deeper organizational characteristics and skill requirements rather than the presence of AI alone.

More broadly, these findings reinforce the view that the labor market effects of AI are heterogeneous and evolving. AI appears to reshape task composition and job design more than it directly alters wage structures, at least in the short run. This underscores the importance of moving beyond exposure-based measures of AI toward understanding the organizational and task-level mechanisms through which technology affects economic outcomes.

8 Conclusion

This study provides new evidence on the role of AI in shaping labor market outcomes using large-scale job posting data. We show that semantic AI relevance is strongly associated with increased job seeker engagement but does not robustly predict higher advertised salaries once standard controls are applied.

By introducing organizational measures of algorithmic control and human agency, we demonstrate that wage variation is not driven by AI exposure itself, but by how work is structured within roles. In particular, human agency emerges as a key determinant of higher compensation, while algorithmic control plays a more limited and unstable role.

These findings highlight an important distinction between attention dynamics and compensation mechanisms in AI-augmented labor markets. While AI signals attract interest and visibility, they do not, on their own, command wage premiums. Instead, the economic value of jobs continues to depend on human-centric capabilities such as judgment, communication, and leadership.

Overall, this paper contributes to the emerging literature on AI and labor economics by providing micro-level, text-based evidence that clarifies how AI affects job demand and compensation differently. It suggests that future research should focus not only on AI exposure, but also on the organizational design of work, as the primary channel through which AI shapes labor market outcomes.

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